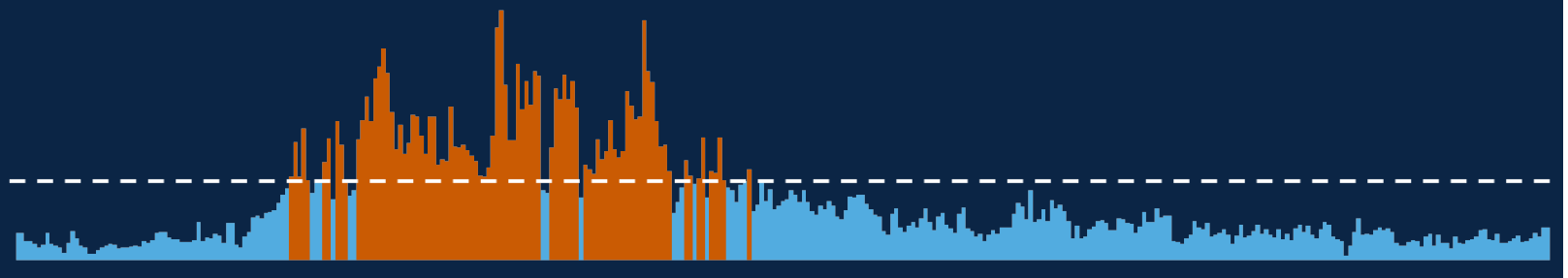
MTH5000 MASTERS RESEARCH PROJECT | ORAL PRESENTATION

Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

Jaykumar Vinodbhai Chauhan
Student ID 34710280 | Master of Mathematics
Supervisor: Dr Tianhai Tian
School of Mathematics, Monash University | November 2026

The last 365 days of the dataset: one bar a day, vermillion above the 121 threshold.



MONASH University

THE FINDING IN TWO NUMBERS

0.379 to 0.881

hit rate at one day ahead as the alarm budget moves from 40 to 100 alarms a year

about 5 points

what changing the model moves it by, at a fixed budget

MOTIVATION

A forecast is only useful if it arrives before the hazardous day does

Health basis. PM2.5 exposure is linked to cardiopulmonary and all-cause mortality (Pope and Dockery 2006; Chen and Hoek 2020).

The event. A day above 121 µg/m^3, the CPCB 'Very Poor' boundary.

The gap. Two Delhi machine learning studies (Masood and Ahmad 2020; Singh and Srivastava 2025) report R^2, RMSE and MAE. Neither evaluates a warning system with an explicit alarm budget.

29.7%

of days exceed 121 µg/m^3 (920 of 3,093)

9.8 yrs

of daily data, Nov 2016 to Aug 2026

1 to 3

days ahead: the forecast horizons

12

model specifications compared

TAKEAWAY **A model with good average error can still miss the extreme days that are the only ones anyone cares about.**

MOTIVATION

Three research questions, and the third turned out to dominate the other two

RQ1 ACCURACY

How accurately can daily concentrations be forecast 1 to 3 days ahead by machine learning, against classical and naive benchmarks?

HEADLINE ANSWER

The classical model (ARIMA) beats every ML regressor at every horizon. The deep networks beat it only at one day ahead.

RQ2 TWO ARMS

Is it better to forecast the concentration and then threshold it, or to classify the exceedance event directly?

HEADLINE ANSWER

Neither dominates: 21 / 11 / 4 across 36 matched comparisons. Forecast-then-threshold is preferred.

RQ3 WARNING SYSTEM

As a warning system, what hit rate and false alarm rate does each approach achieve, and how does the threshold trade them off?

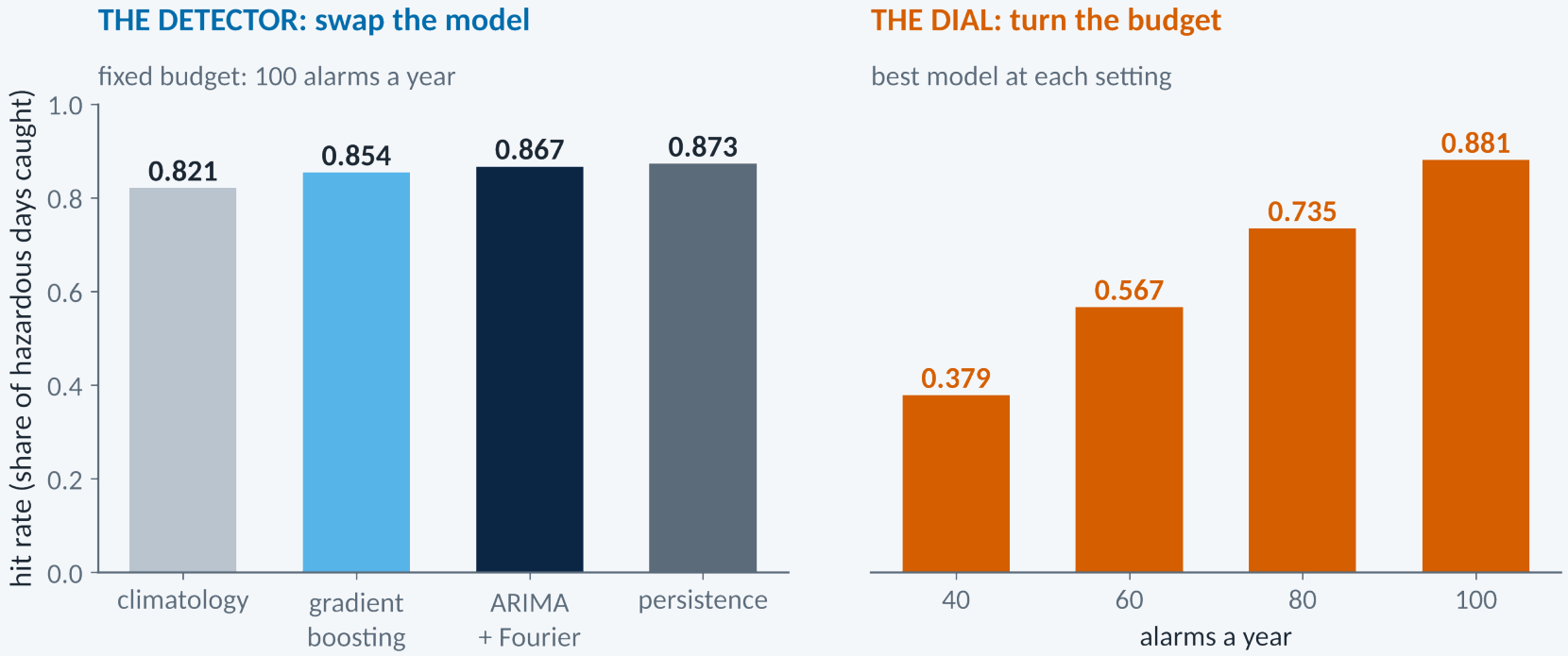
HEADLINE ANSWER

The alarm budget moves hit rate by 50 points. The model moves it by about 5.

TAKEAWAY

The third question turns out to dominate the other two, and the talk is organised around it.

The idea in one picture: the detector matters little, the dial matters a lot



Hit rate at one day ahead (Table 4.7). Hit rate: the share of hazardous days that were caught.

TAKEAWAY

Like a smoke alarm: swapping the brand moves the hit rate from 0.82 to 0.87; turning the sensitivity dial moves it from 0.38 to 0.88.

The route: from the data, through the models, to the alarm budget

1

Data

about 7.25 min

one station, two defects, no weekly cycle

2

Methods

about 5.5 min

twelve models, rolling-origin tests, warning measures

3

Results

about 11.5 min

RQ1 accuracy, RQ2 two arms, RQ3 the alarm budget

4

Discussion

about 1.75 min

robustness checks and limits

5

Conclusion

about 3.25 min

contributions and the recommendation

How to read the colours: data and models · ARIMA, the benchmark · the number to remember

TAKEAWAY **Hold one question in mind: how much does the model matter, compared with the number of alarms you allow?**

One well-characterised Delhi monitor, nearly ten years of daily PM2.5

Quantity	Value after cleaning
Source	OpenAQ location 8118, sensor 23534, AirNow
Period	2016-11-09 to 2026-08-27
Calendar days	3,579
Usable daily means	3,093 (86.4% of calendar)
Longest gap	34 consecutive days
Mean, median, maximum	102.7, 71.7, 896.0 µg/m^3
Days above 121	920 (29.7%)

Report Table 3.1.

Why station 8118, and what it costs

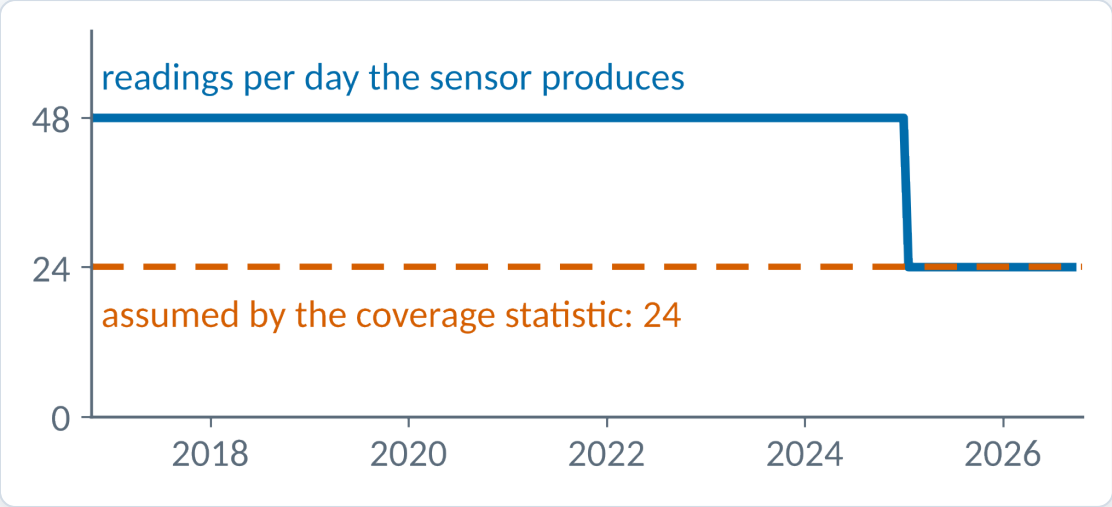
- Five stations compared on coverage, not nominal record length.
- Every CPCB station serves data only from 19 Feb 2025 (about 18 months).
- Only 8118 returns the full record.

Cost: an AirNow feed, almost certainly a US diplomatic post monitor. Results describe one site.

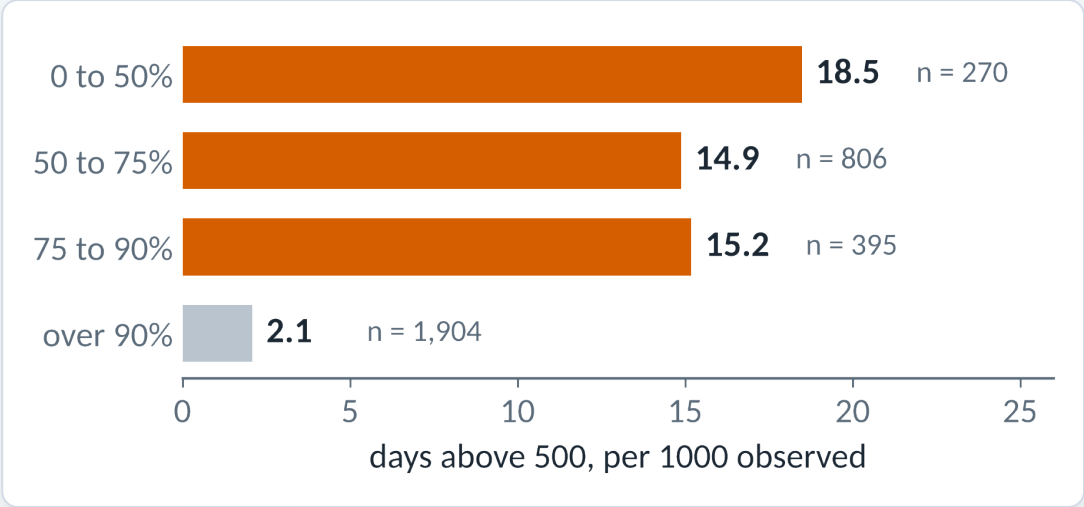
TAKEAWAY **One reference site with a full decade of record: good for comparing methods, not a claim about all of Delhi.**

Two defects in the source data were found and corrected before any model was fitted

(a) What the sensor really reports



(b) Implausible values, by true coverage



A clipped ceiling. 1990 appears three times identically, once on 2 September, in the monsoon, between days of 42 and 178. Five negatives too.

A wrong denominator. Completeness assumed 24 readings a day. 1,085 of 3,467 days are below 75% true coverage; only 268 have no value at all.

TAKEAWAY

Rules: below 0 (5 days), coverage below 50% (270), above 1000 (7): 282 of 3,375 removed, and the exceedance rate moves only 30.0% to 27.0%.

Exceedances arrive as multi-day winter episodes, and that is the structure worth forecasting

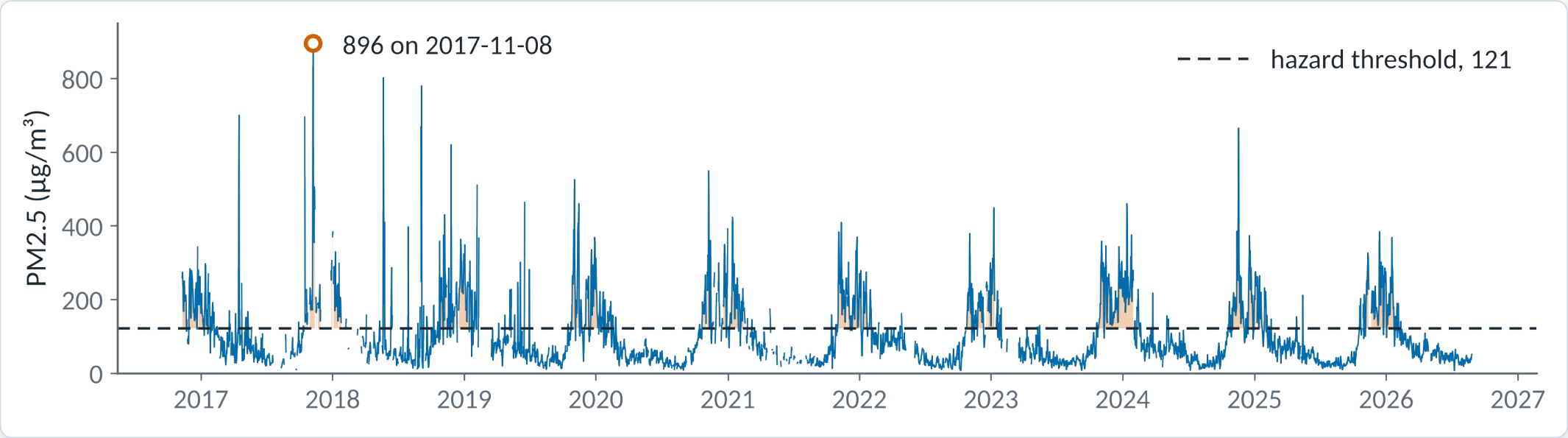
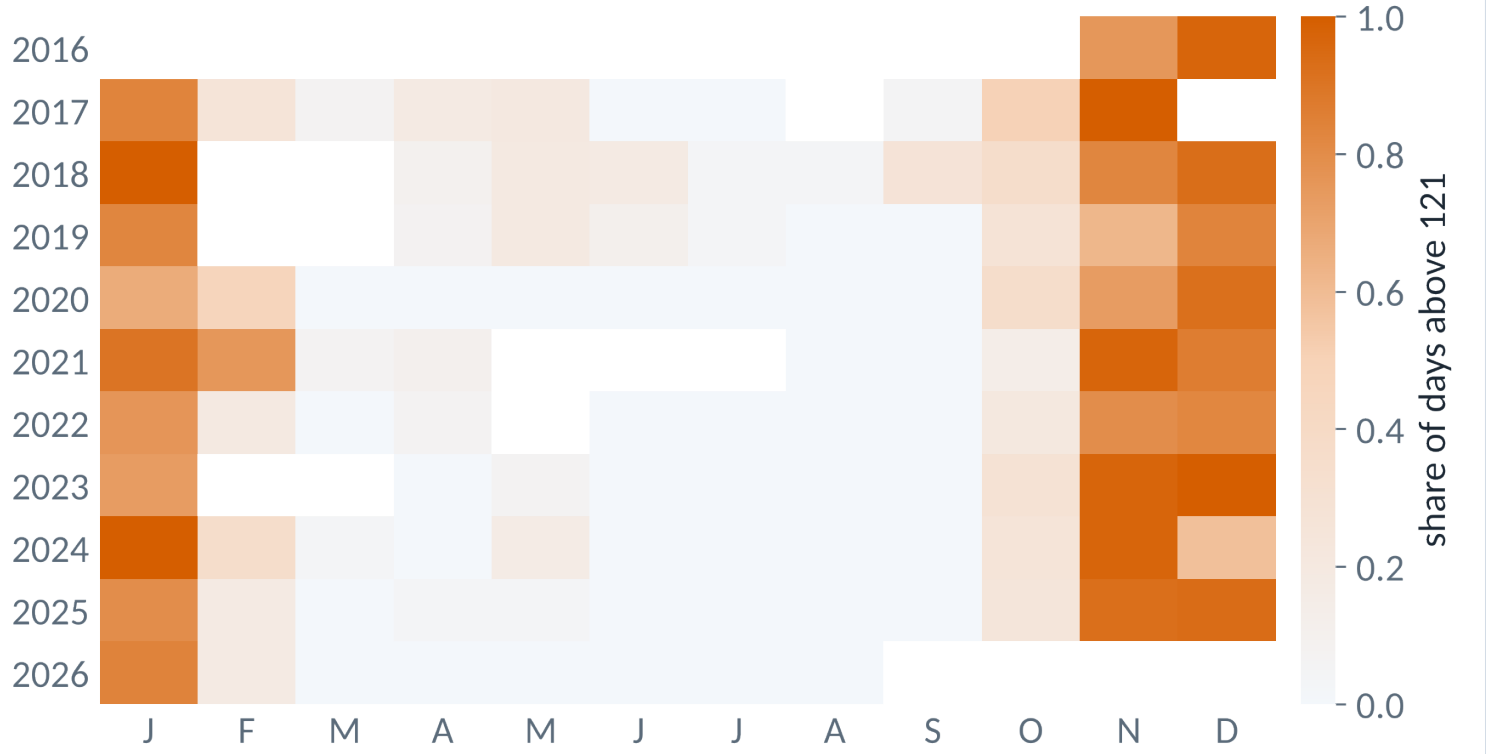


Figure 3.2. The cleaned daily series; days above the hazard threshold are shaded. Maximum 896.0 on 2017-11-08.

102.7	mean µg/m³
71.7	median
2.17	skewness
8.31	excess kurtosis

TAKEAWAY **Episodes make persistence a strong forecast, and the heavy right tail is why every model is fitted to log PM2.5.**

Every year, the same months: hazardous days gather from October to February



Each cell is a month; the darker, the larger the share of its days above 121.

The pattern repeats every year: October to February is dark; the monsoon months are almost clear.

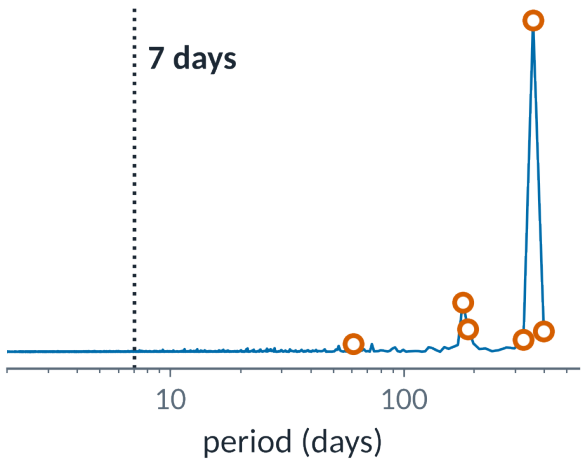
That regularity is the annual cycle the models carry, and why a calendar-only forecast (climatology) is a strong baseline.

TAKEAWAY

Hazard is seasonal and arrives in episodes, so the calendar alone is already a strong baseline that a model has to beat.

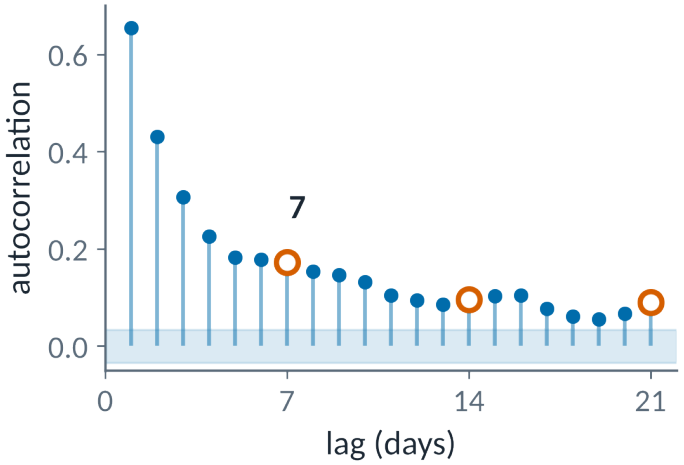
An assumption from the approved proposal fails: there is no weekly cycle at this station

Periodogram



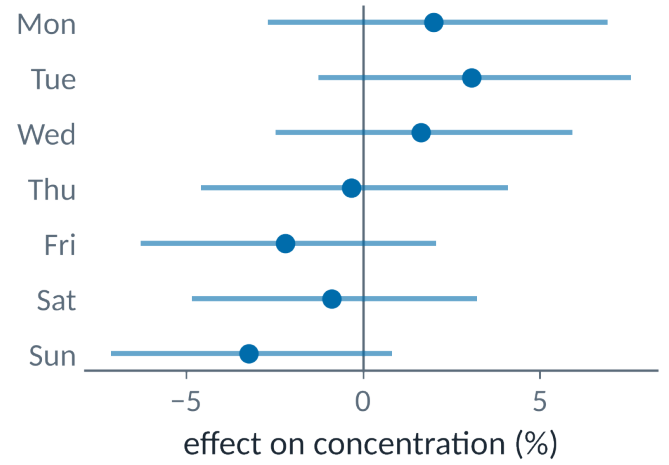
Dominant periods 357.9, 178.9, 188.4, 397.7, 325.4, 60.7 days. None near 7.

Autocorrelation



Lag 7 = 0.173, between lag 6 (0.179) and lag 8 (0.153): on the decay, not above it.

Day of week

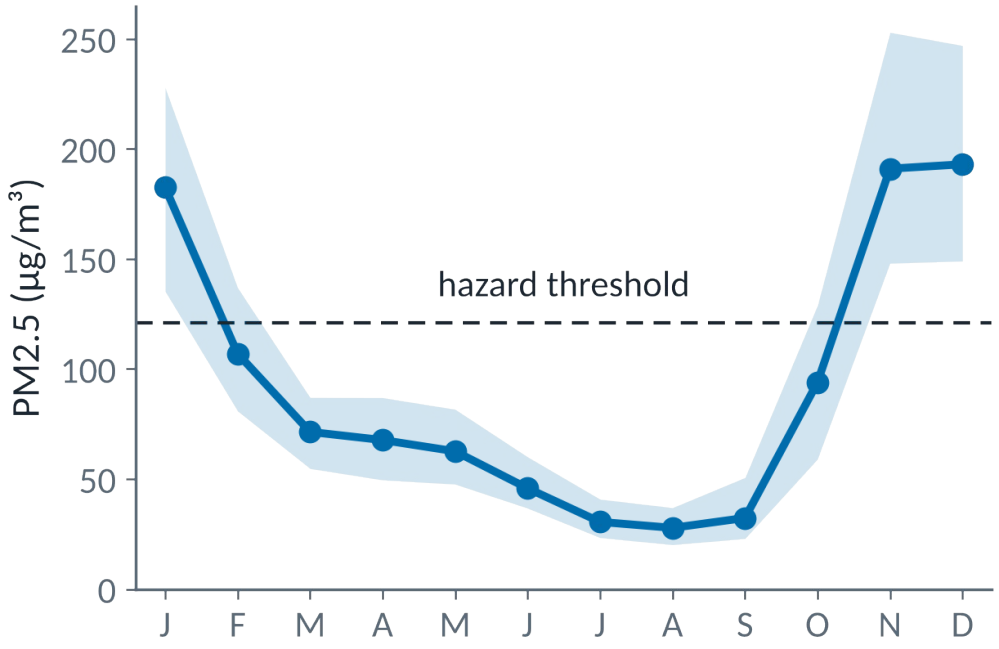


6.5% spread. ANOVA F = 1.13, p = 0.34. Kruskal-Wallis p = 0.77.

TAKEAWAY

Three independent tests agree: ARIMA carries the annual cycle only, with no weekly seasonal order.

The annual cycle is unmistakable, so ARIMA carries it with Fourier terms and no weekly order



$$(1 - \phi_1 B)(1 - B)\, y_t = c + (1 + \theta_1 B)\, \varepsilon_t + \Sigma_{k=1}^{4} [\, \alpha_k \sin(2\pi k t/365.25) + \beta_k \cos(2\pi k t/365.25) \,]$$

y_t = log(PM2.5), B is the backshift operator, K = 4 annual Fourier pairs.

Order by AIC over a 24-candidate grid: (1,1,1) at 1892.5, next (1,1,2) at 1893.6.

Fitted: $\phi_1 = 0.581$ and $\theta_1 = -0.987$, which is 0.013 inside the invertibility boundary.

Back-transform: median exp(m); mean exp(m + $s^2/2$). Both reported.

Figure 3.5. Monthly median with interquartile range.

TAKEAWAY

A simpler, better-identified specification: the data justified dropping the weekly term, and the Fourier terms carry the season.

METHODS

Twelve specifications from three families, all scored by one protocol

Classical and naive

- Persistence (repeat today)
- Climatology
- Seasonal naive, lag 7
- ARIMA(1,1,1) + annual Fourier terms

Machine learning

- Ridge regression
- Random forest: level and change
- Histogram gradient boosting: level and change

Deep learning, 10 seeds each

- GRU
- LSTM
- Transformer encoder
- Kolmogorov-Arnold-LSTM: bounded check only

TAKEAWAY

Fitted to log PM2.5, forecast at h = 1, 2, 3 days, refitted as the training window expands, scored on identical origins by identical code.

Three regressors, each controlling variance by a different mechanism

Ridge regression

SHRINKAGE

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

- Closed form: unique for any $\lambda > 0$, even with collinear lags.
- λ by cross-validation at every refit.

Random forest

BAGGING

$$\operatorname{Var} \hat{f}(x) = \rho\sigma^2 + (1 - \rho)\sigma^2 / B$$

- Average of B trees on bootstrap resamples.
- Random feature subsets lower ρ, the floor averaging cannot beat.

Histogram gradient boosting

ADDITIVE CORRECTION

$$F_m = F_{m-1} + \nu\, h_m$$

- Each tree fits the pseudo-residual: descent in function space.
- Binned features accept missing values, so nothing is imputed.

TAKEAWAY **Trees are fitted to both the level and the change from the origin, because a tree cannot extrapolate and this series drifts.**

Three networks under one protocol, so the comparison is the cell and nothing else

Architecture	Size	Optimiser	Parameters
GRU	24 hidden units	Adam, lr = 0.01	2,329
LSTM	24 hidden units	Adam, lr = 0.01	3,097
Transformer	d_model = 16, 2 heads, 1 layer	Adam, lr = 0.001	3,409
Kolmogorov-Arnold-LSTM	24 hidden units, grid size 5	Adam, lr = 0.01	3,288

Gates

GRU: 2

LSTM: 3 + cell state

Transformer: self-attention

Report Table 3.6. Parameters are for the six-channel network, confirmed against the fitted model objects.

Shared input: last 30 days; log PM2.5, a missingness flag, two annual Fourier pairs.

Shared training: target is the change in log concentration; Adam on absolute error; early stopping on the latest 15%.

Reported as a range over ten seeds, never one favourable seed.

TAKEAWAY **Only the cell differs: gated recurrence, a memory cell, or attention. Each is reported as a ten-seed range.**

Rolling-origin evaluation: 2,303 forecasts, none of which can see the future



Invariant: at origin t, observations up to and including day t, and nothing else.

Tested, not assumed: corrupt the series from a chosen date, rerun, and require no earlier forecast to change.

One model per horizon: the label on row t′ is the value at t′ + h, so training ends at t − h.

Why not one train-test split?

0.903 → 0.940

single block — rolling origin

ARIMA rMAE at one day. The single block was calmer and flattered every model.

TAKEAWAY **The protocol is the credibility of every number that follows: leakage-tested, and one draw of history is never trusted.**

Scored as a warning system: an alarm budget in alarms per year, not just an error

ACCURACY

$$\text{rMAE} = \text{MAE(model)} / \text{MAE(persistence)}$$

1.000 is persistence; below 1 beats it. Per horizon, never averaged.

HIT RATE

$$\text{TP} / (\text{TP} + \text{FN})$$

The share of actual hazardous days the model catches.

FALSE ALARM RATE

$$\text{FP} / (\text{FP} + \text{TN})$$

False alarms divided by quiet days, not by alarms raised.

ALARMS PER YEAR

$$365.25\ (\text{TP} + \text{FP}) / n$$

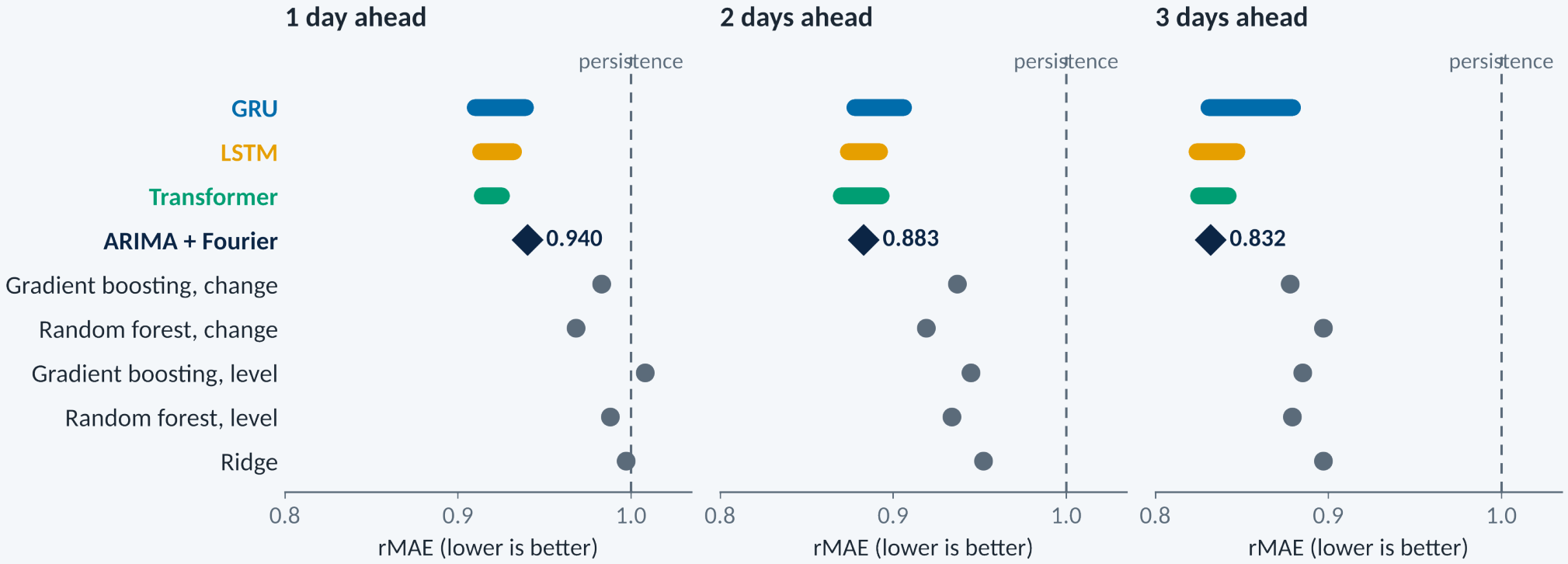
The unit an operator budgets in.

TAKEAWAY

Plain accuracy is never reported: with hazardous days a minority, a model that never alarms scores high by construction.

RQ1: the networks win only at one day ahead, and ARIMA beats every ML regressor

rMAE: error compared with simply repeating yesterday's value (the dashed line at 1.0). Further left is better.



Report Table 4.1 drawn: bars are ranges over ten seeds; the diamond is ARIMA; dots are single fits; the dashed line is persistence (rMAE = 1).

TAKEAWAY

Networks: 0.910 to 0.939 vs ARIMA 0.940 at one day. At three days 0.831 vs 0.832 is a tie. Every ML regressor trails ARIMA throughout.

Both models follow each winter episode; the networks' edge shows on the sharpest rises

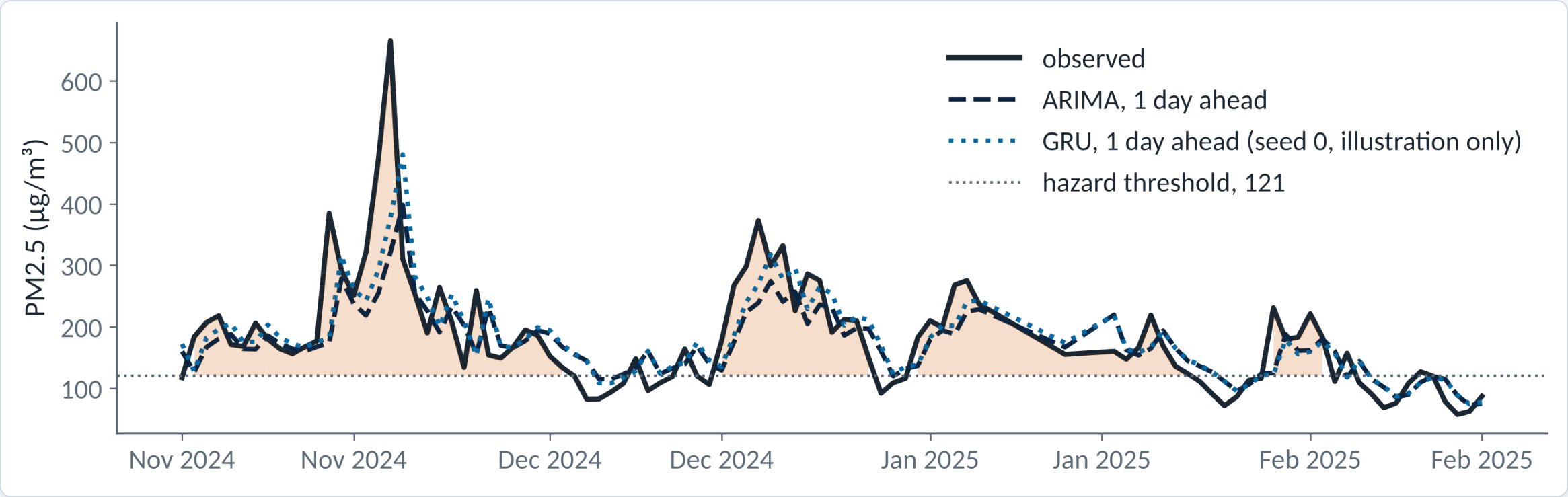
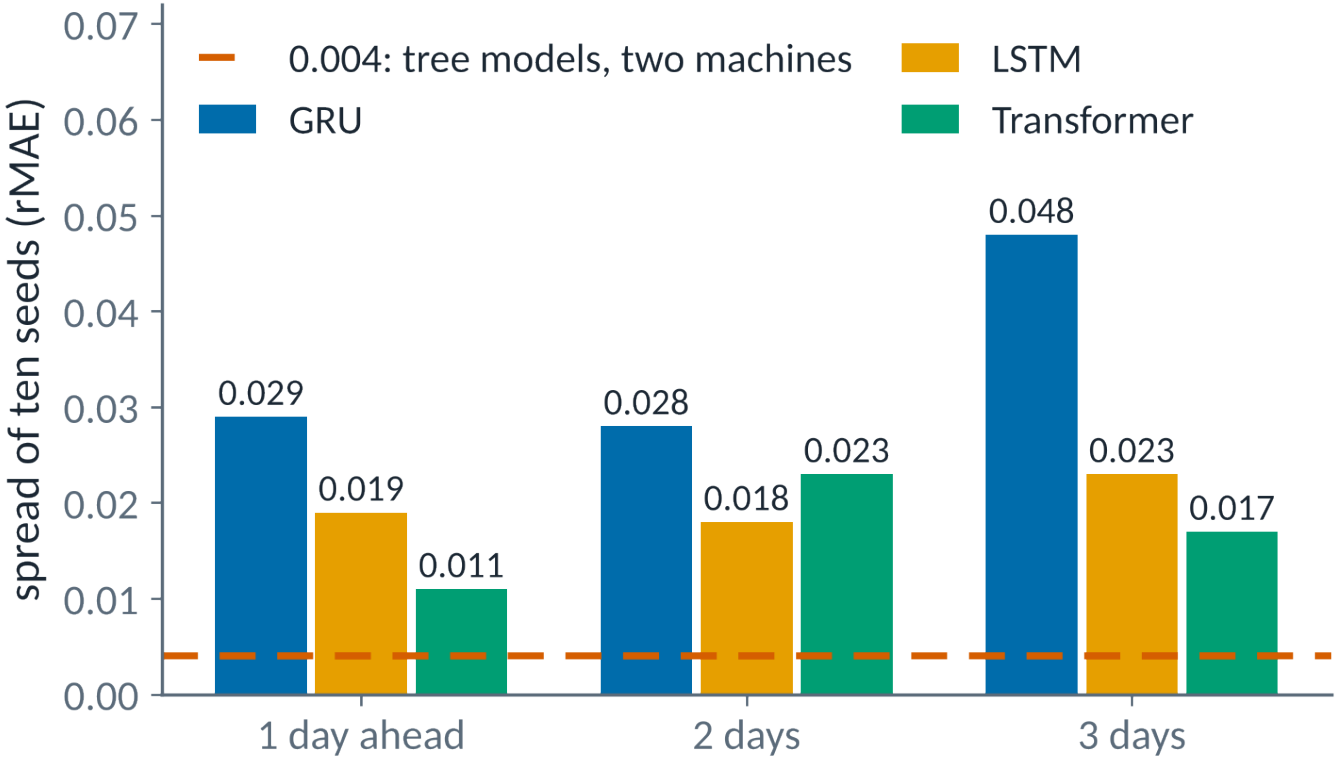


Figure 4.2. One representative winter, November 2024 to February 2025. The GRU line is seed 0 alone, for illustration only.

TAKEAWAY **The edge is a short-range one: the forecast tracks the shape of each episode, with the gap on the sharpest rises.**

A network's result is a range, not a number, and the spread rivals the gaps between models



Spread of ten seeds: GRU 0.029, 0.028, 0.048 at h = 1, 2, 3; wider than some gaps in the table.

Noise floor 0.004: tree models moved this much between two machines.

Same seed, same machine still moved the GRU nearly as much as a new seed.

So five seeds became ten.

TAKEAWAY **A single seed could be reported as beating or losing to a rival at the analyst's convenience: only ranges are honest.**

RESULTS

Formal tests are more conservative than counting seeds, and the picture holds

Seeds with p < 0.05 vs ARIMA	h = 1	h = 2	h = 3
GRU	**4 / 10**	0 / 10	1 / 10
GRU with weather	**7 / 10**	0 / 10	1 / 10
LSTM	**5 / 10**	0 / 10	0 / 10
Transformer	**7 / 10**	0 / 10	0 / 10

Report Table 4.2. The one significant 3-day GRU seed (p = 0.0056) favours ARIMA.

Four of ten significant is a small-sample proportion: 95% Wilson interval 0.17 to 0.69.

14 of 15

regressor tests favour ARIMA at p < 0.05

6 of 27

tests survive a Bonferroni correction

Read beyond one day as suggestive

Twenty-seven tests, no joint correction: about 1.4 false positives expected by chance. After Bonferroni, none of the plain GRU, LSTM or transformer survives.

TAKEAWAY **One-day wins are real but modest; beyond one day nothing is confirmed, and the report says so.**

Two reasons for ruling models out did not survive being tested

ARGUMENT 1: FEWER PARAMETERS FAVOUR THE GRU

LSTM beats GRU at every horizon

by 0.005, 0.008, 0.013 rMAE

3,097 parameters against 2,329, a third more. The smaller network is not winning because it is smaller.

ARGUMENT 2: TOO LITTLE DATA FOR A TRANSFORMER

Transformer beats GRU in 8 of 10 seeds

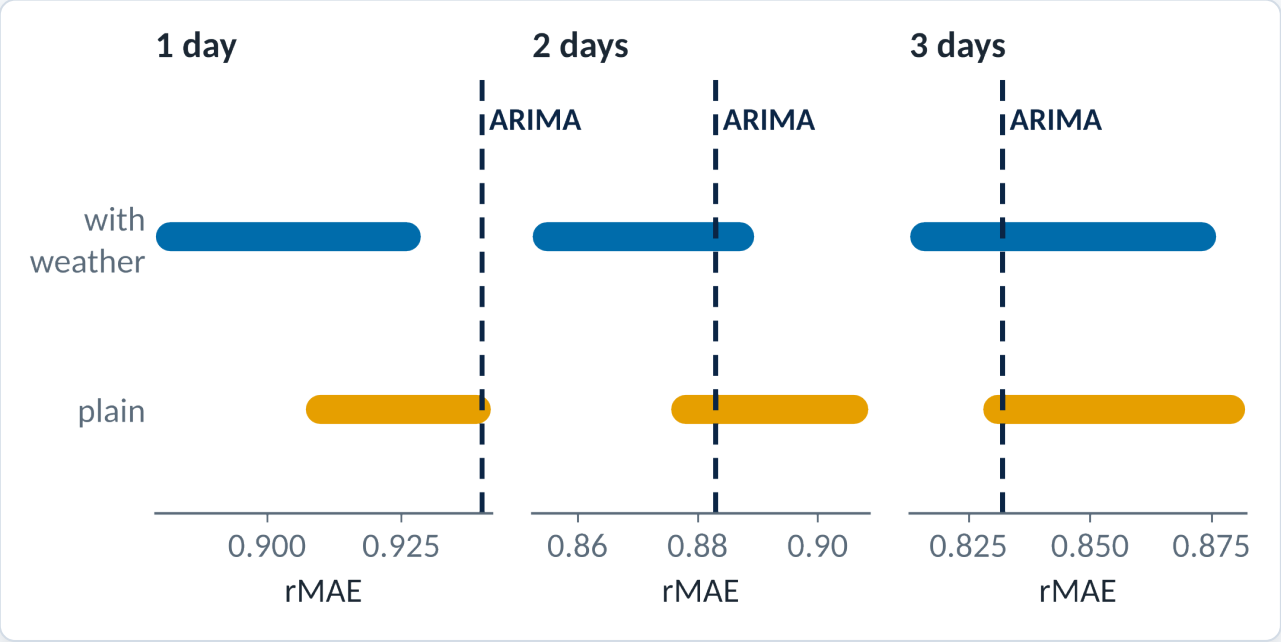
at every horizon, and matches the LSTM

Dr Tian's count of about 4,000 days clears the threshold. At one day it has the strongest formal result: 7 of 10 seeds significant.

TAKEAWAY

A defensible prior is not evidence: both arguments were tested directly and neither is supported. The transformer is one untuned configuration.

Weather helps at one day ahead and leaves two and three days a tie



GRU ranges over ten seeds. Plain from Table 4.1, with weather from Table 4.4. Dashed: ARIMA.

Change in rMAE	h = 1	h = 2	h = 3
Gradient boosting, level	−0.082	−0.020	−0.029
Gradient boosting, change	−0.073	−0.047	+0.006
Ridge	−0.031	−0.005	+0.023
Random forest, level	+0.002	−0.006	+0.053
Random forest, change	−0.003	−0.001	+0.022

Report Table 4.4: change from adding weather. Negative is an improvement.

TAKEAWAY

GRU with weather beats ARIMA at one day in all ten seeds. At two days: 3 win, 6 tie, 1 loses. At three: 3 win, 2 tie, 5 lose.

RESULTS

Carried through the decision threshold, a more accurate forecaster buys about one point of hit rate

Best hit rate, alarms per year	40	60	80	100
h = 1, no weather	0.401	0.609	0.787	0.899
h = 1, weather models added	0.402	0.609	0.792	**0.909**
h = 2, no weather	0.388	0.597	0.771	0.900
h = 2, weather models added	**0.400**	0.597	0.771	0.900
h = 3, no weather	0.383	0.567	0.756	0.876
h = 3, weather models added	0.383	**0.577**	0.756	0.876

Report Table 4.5: best hit rate within an alarm budget, averaged over the ten network seeds, on identical origins.

≤ 1.2 points

the largest hit-rate gain from adding weather models

8.2 points

rMAE gain for gradient boosting at h = 1, which became at most a little over 1 point of hit rate

TAKEAWAY **The ceiling is not the models being too inaccurate: it persists after a materially more accurate forecaster is added.**

RESULTS

RQ2: forecast-then-threshold has a real but modest edge over direct classification

h = 1, alarms per year	40	60	80	100
Ridge: forecast then threshold	**0.377**	**0.554**	**0.723**	0.850
Ridge: penalised logistic	0.356	0.550	0.721	**0.856**
Random forest (change): forecast	**0.367**	0.550	**0.729**	**0.867**
Random forest: classifier	0.331	0.550	0.719	0.856
Grad. boosting (change): forecast	**0.379**	**0.554**	0.727	0.854
Grad. boosting: classifier	0.323	0.546	0.727	**0.865**

Report Table 4.6: matched by base algorithm, h = 1. Shaded: the better arm; exact ties unshaded.

21 forecast-then-threshold wins

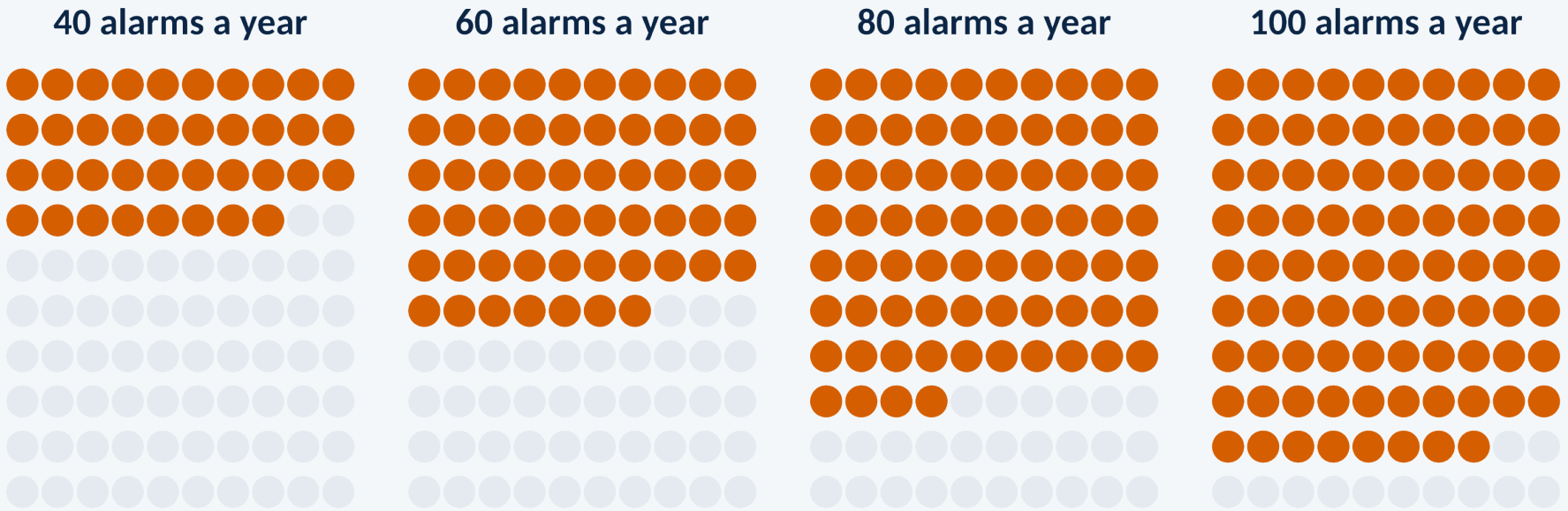
11 direct classification wins

4 exact ties

of 36 matched comparisons

TAKEAWAY **Median gap 0.011 of hit rate. Forecast-then-threshold wins 8 of 9 at a 40-alarm budget, but loses 6 of 9 at 100.**

What an alarm budget means: of every 100 hazardous days, how many are caught

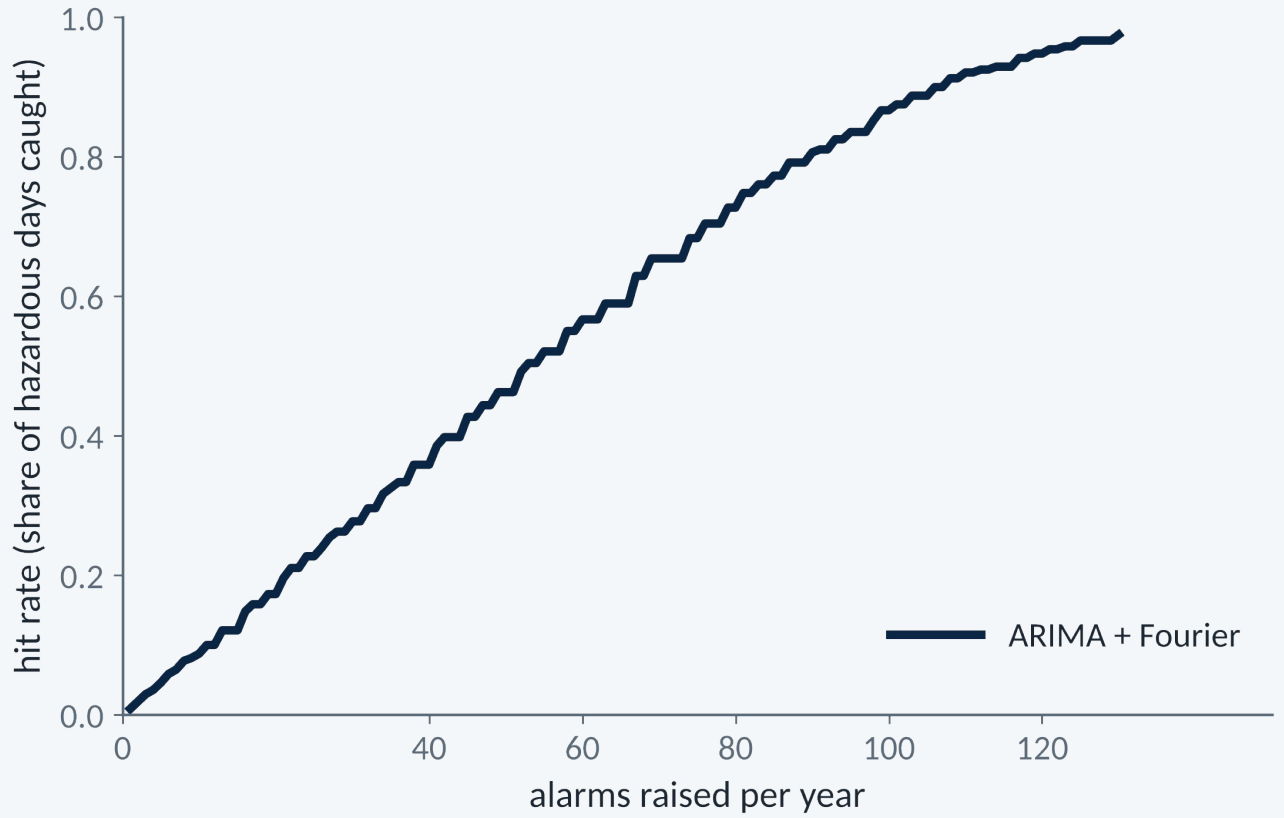


Best model at each budget, one day ahead (Table 4.7), rounded to whole days out of 100.

TAKEAWAY

From one alarm every nine days to one every four, the same model catches 88 of 100 hazardous days instead of 38: a policy choice, not a modelling one.

RQ3, step 1: one model, and its hit rate rises as you allow more alarms



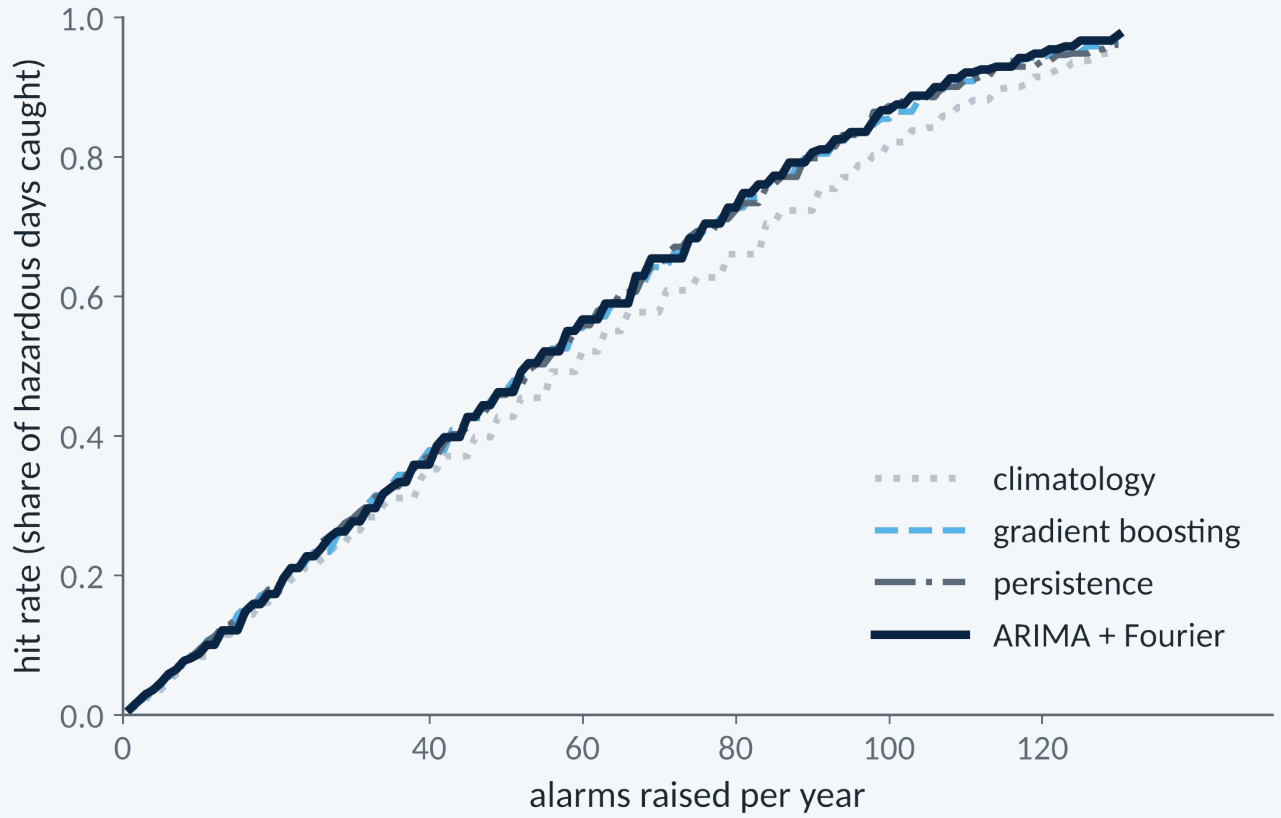
Horizontal: alarms raised per year, the dial.

Vertical: the share of hazardous days caught.

More alarms catch more days, by construction, so this alone says nothing about models.

TAKEAWAY **Allow more alarms and the same model catches more hazardous days: that is the dial.**

RQ3, step 2: add three very different models, and the curves nearly coincide



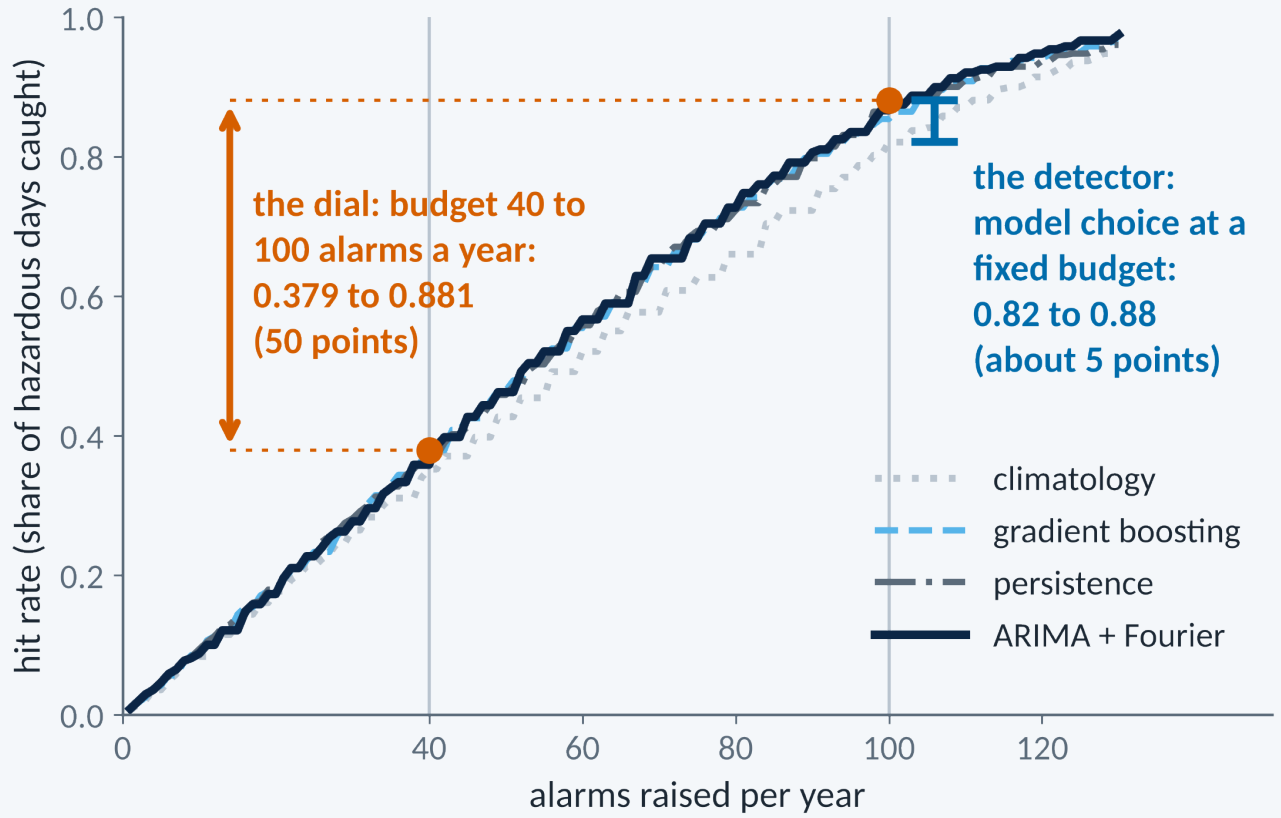
ARIMA, gradient boosting, persistence (just repeat yesterday) and climatology (the calendar alone).

Four very different forecasters, almost one curve.

Deliberately not separated: the overlap is the finding.

TAKEAWAY **Swap the detector, and the curve barely moves.**

RQ3: moving the alarm budget changes the hit rate by fifty points, and changing the model by five



50 points

the dial: budget 40 to 100 alarms a year (h = 1)

about 5 points

the detector: model choice at a fixed budget

At h = 3 and 40 alarms a year, persistence (0.359) is the best model in the whole study.

TAKEAWAY

The dial matters more than the detector: the alarm budget is the policy instrument, and the model only supplies its input.

The central finding survives every check, and the checks changed what I can claim

Check	What changed	Result
Restore the 7 capped days	Refit full harness on the uncapped series	ARIMA 0.935 vs 0.940 (h = 1). Ridge 0.997 → 1.118. GB level 0.883 → 1.006 (h = 3).
Weather coverage gap	Restrict to origins inside the observed window	One-day ordering survives; two- and three-day picture unchanged.
Seed count 5 → 10	Doubled for every network	Weather win at h = 2 and 3 did not survive.
GRU window and size (bounded)	6 settings, 3 seeds, one split	Default near best at h = 1, 2; worst of six at h = 3 (45-day window 2.2 points better).
GRU Fourier pairs (bounded)	2 pairs → 4, as in ARIMA	h = 3 improves 3.96 points, enough to put the mean below ARIMA on this split.
GRU optimiser (bounded)	Adam, RMSprop, SGD; lr ×/÷ 3	h = 3: SGD with momentum about 2 points better (0.839 vs 0.858).

TAKEAWAY **Bounded checks are single-split, three-seed: nothing in the ten-seed Table 4.1 was changed on their strength.**

DISCUSSION

What the results do not show

One site, one city

8118 is a diplomatic-enclave monitor. Nothing here is a claim about Delhi as a whole or about other cities.

Observed weather, not forecast

A real system would use numerical weather prediction. Boundary layer height is not available from meteostat.

Scope boundaries

LSTM and transformer: no weather channels, no threshold sweep. The KAN-LSTM is a bounded check only.

Small search spaces

Tree grids were small, so ML results are a lower bound. The GRU's 3-day configuration is the one its own checks least defend.

Significance testing

27 tests, no joint correction, best-seed selection, no bootstrap of the loss differential's heavy tails.

Not bit-reproducible

Network training moves at a fixed seed; ten seeds reduce the problem but do not remove it.

TAKEAWAY

None of these overturns the central finding: it rests on several independent sources agreeing, not on any single p-value.

What this project contributes

1

A warning-system evaluation

Hit rate, false alarm rate and alarms per year under an explicit budget: the operating point the Delhi studies do not report.

2

A leakage-tested comparison

Twelve specifications, classical to deep, under rolling-origin evaluation, with networks as ten-seed ranges.

3

Two data defects, diagnosed and fixed

A clipped ceiling and a coverage statistic that assumed the wrong cadence, both corrected before modelling.

4

Priors tested, not assumed

No weekly cycle; the GRU is not better because it is smaller; the small-data case against a transformer fails.

TAKEAWAY

Applying a standard model suite to Delhi is not new; evaluating it as a warning system with a varying alarm budget is.

CONCLUSION

Three questions asked, three answered

RQ1	**Accuracy vs benchmarks**	ARIMA beats every ML regressor at every horizon (rMAE 0.940 at h = 1). The GRU beats ARIMA only at h = 1, across all ten seeds. The LSTM and transformer follow the same pattern.
RQ2	**Forecast then threshold, or classify?**	Neither dominates: 21 / 11 / 4 across 36 matched comparisons. Forecast-then-threshold is preferred, since a concentration can be rethresholded for any future policy.
RQ3	**Model choice vs decision threshold**	The budget sweep moves hit rate from 0.379 to 0.881 (h = 1, 40 to 100 alarms a year). Model choice moves it about 5 points at a fixed budget.

TAKEAWAY

Machine learning earns a real but narrow place; the dial (the alarm budget), not the detector (the model), decides how many hazardous days are caught.

Design the warning system around its alarm budget, and treat the model as a refinement

RECOMMENDATION

Design a hazardous air quality warning system around the alarm budget its operator can justify, with the forecasting model a secondary refinement.

A model comparison that does not fix the alarm budget it was evaluated at has answered a question nobody who operates the system asked.

SIX EXTENSIONS

1. **Forecast weather.** Numerical weather prediction in place of observed conditions.
2. **Boundary layer height.** From a reanalysis such as ERA5.
3. **Four harmonics for the GRU** under the full ten-seed evaluation.
4. **A second station,** ideally inside the CPCB network.
5. **Complete the LSTM and transformer:** weather channels and the threshold sweep.
6. **Promote the KAN-LSTM** from a bounded check to ten seeds.

Thank you. Questions?

The alarm budget a decision maker accepts dominates the choice of model.

Code, data and report: github.com/JC16092/mth5000-aqi-forecast

Backup: B1 full Table 4.1, B2 ARIMA order search, B3 significance tables, B4 GRU sensitivity, B5 stability and architecture checks, B6 RQ2 at all horizons, B7 full Table 4.7, B8 to B9 likely questions, B10 plain-language glossary.

B1. Report Table 4.1 in full: rMAE against persistence

Model	h = 1	h = 2	h = 3	Bias, h = 1
GRU, 10 seeds	0.910 to 0.939	0.878 to 0.906	0.831 to 0.879	−2.8
LSTM, 10 seeds	0.913 to 0.932	0.874 to 0.892	0.824 to 0.847	−2.4
Transformer, 10 seeds	0.914 to 0.925	0.870 to 0.893	0.825 to 0.842	−2.8
ARIMA(1,1,1) + Fourier	**0.940**	**0.883**	**0.832**	**−6.4**
Random forest, change	0.968	0.919	0.897	+2.1
Gradient boosting, change	0.983	0.937	0.878	+2.0
Random forest, level	0.988	0.934	0.879	+4.1
Gradient boosting, level	1.008	0.945	0.885	+4.0
Ridge regression	0.997	0.952	0.897	+2.6
Persistence	1.000	1.000	1.000	−0.1
Climatology	1.415	1.060	0.947	+9.4
Seasonal naive, lag 7	1.759	1.325	1.168	+0.1

Over 2,303 origins. Lower is better; 1.000 is persistence.

BACKUP

B2. The ARIMA order search in full: all 24 candidates by AIC

(p,d,q)	AIC	BIC	note
(1,1,1)	**1892.5**	**1955.8**	**chosen**
(1,1,2)	1893.6	1962.6	
(2,1,1)	1894.4	1963.4	
(3,1,1)	1895.8	1970.5	
(2,1,2)	1895.8	1970.6	no convergence
(3,1,2)	1897.7	1978.2	
(1,0,2)	1910.4	1985.2	no convergence
(2,0,1)	1912.0	1986.7	no convergence
(1,0,1)	1912.1	1981.1	
(2,0,0)	1914.7	1983.7	
(3,0,1)	1915.5	1996.0	no convergence
(3,0,2)	1919.9	2006.1	

(p,d,q)	AIC	BIC	note
(1,0,0)	1927.4	1990.7	
(0,0,2)	2009.2	2078.2	
(0,1,2)	2022.5	2085.7	
(2,0,2)	2100.0	2180.5	no convergence
(3,0,0)	2126.5	2201.2	no convergence
(3,1,0)	2153.8	2222.8	no convergence
(0,0,1)	2167.9	2231.1	
(0,1,1)	2177.9	2235.4	
(2,1,0)	2196.7	2259.9	no convergence
(1,1,0)	2265.2	2322.7	
(0,1,0)	2355.6	2407.4	
(0,0,0)	2734.0	2791.5	

Report Table 3.4, sorted by AIC.

BACKUP

B3. Diebold-Mariano tests against ARIMA

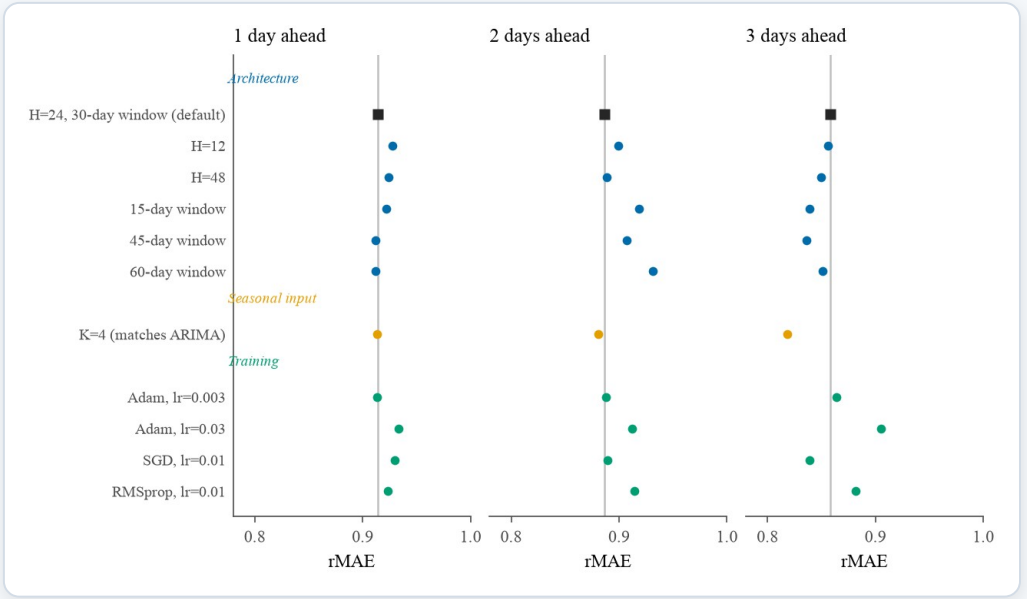
Network vs. ARIMA	h	DM statistic	p-value	Sig. seeds
Plain GRU	1	−3.08 to −0.10	0.002 to 0.923	4/10
Plain GRU	2	−0.47 to 1.73	0.083 to 0.992	0/10
Plain GRU	3	−0.05 to 2.77	0.006 to 0.987	1/10
GRU with weather	1	−3.92 to −0.87	<0.001 to 0.383	7/10
GRU with weather	2	−1.78 to 0.26	0.075 to 0.942	0/10
GRU with weather	3	−0.83 to 1.97	0.049 to 0.907	1/10
LSTM	1	−2.59 to −0.66	0.010 to 0.511	5/10
LSTM	2	−0.65 to 0.83	0.409 to 0.974	0/10
LSTM	3	−0.59 to 1.05	0.292 to 0.995	0/10
Transformer	1	−2.74 to −1.39	0.006 to 0.164	7/10
Transformer	2	−1.34 to 0.84	0.181 to 0.995	0/10
Transformer	3	−0.63 to 0.73	0.463 to 0.988	0/10

ARIMA vs. regressor	h = 1	h = 2	h = 3
Ridge	4.25 (<0.001)	3.85 (<0.001)	3.62 (<0.001)
RF, level	3.05 (0.002)	2.43 (0.015)	2.19 (0.029)
RF, change	1.99 (0.046)	**1.93 (0.054)**	3.53 (<0.001)
GB, level	3.92 (<0.001)	2.65 (0.008)	2.26 (0.024)
GB, change	3.00 (0.003)	2.51 (0.012)	1.96 (0.050)

Report Tables 4.2 and 4.3. Cells show DM statistic (p-value). Shaded: the one non-significant regressor pairing, p = 0.054.

B4. GRU sensitivity: three bounded checks, three seeds, one split

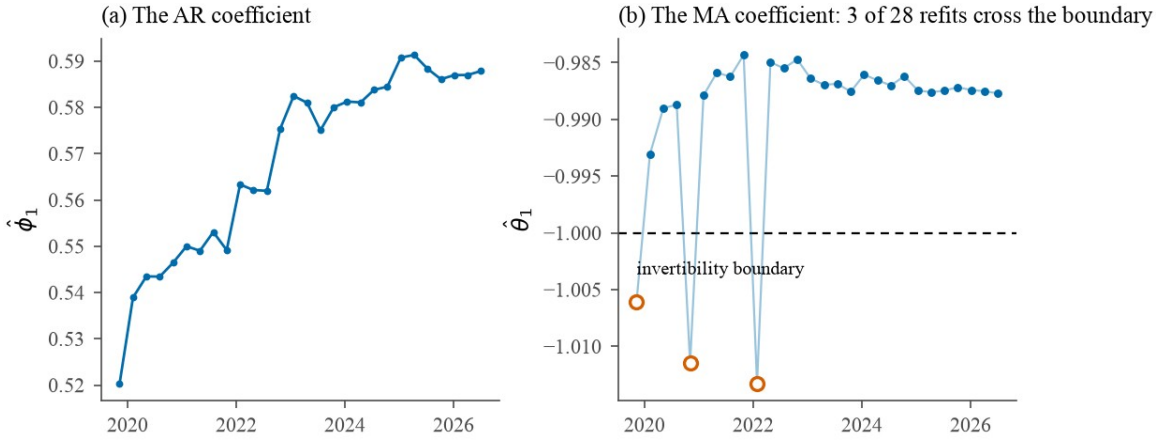
Check	Setting	h = 1	h = 2	h = 3
Architecture	H = 24, 30-day (default)	0.914	0.886	0.858
	H = 12	0.927	0.899	0.856
	H = 48	0.924	0.889	0.850
	15-day window	0.922	0.919	0.839
	45-day window	0.912	0.907	0.836
	60-day window	0.912	0.931	0.851
Seasonal input	K = 4 (matches ARIMA)	0.913	0.881	0.819
Training	Adam, lr = 0.003	0.913	0.888	0.864
	Adam, lr = 0.03	0.933	0.912	0.905
	SGD, lr = 0.01	0.930	0.889	0.839
	RMSprop, lr = 0.01	0.923	0.914	0.882



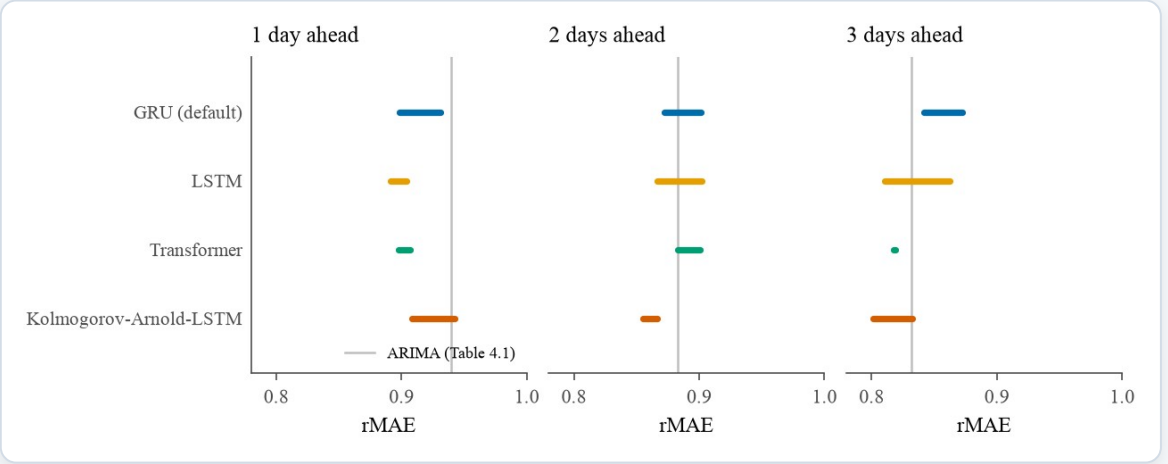
Report Figure 5.1. The vertical line marks the default.

BACKUP

B5. ARIMA coefficient stability, and all four architectures' bounded checks



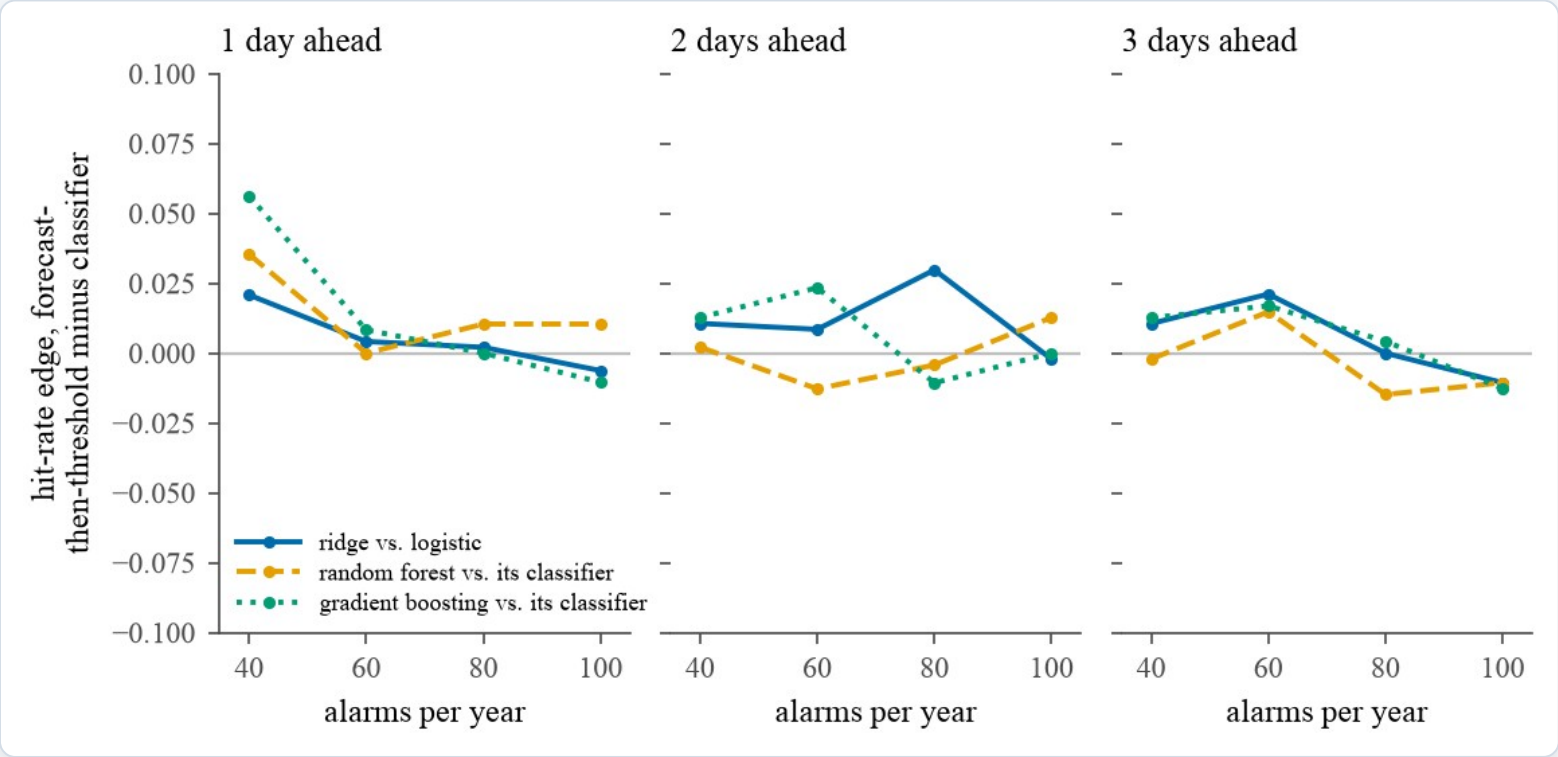
Report Figure 3.6. The three open points cross the invertibility boundary.



Report Figure 5.2. Bounded checks, ranges over three seeds, against ARIMA's rolling-origin rMAE.

BACKUP

B6. RQ2 at all three horizons: the hit-rate advantage of forecast-then-threshold



Report Figure 4.4. One line per matched pair.

TAKEAWAY

Forecast-then-threshold wins 8 of 12 matched comparisons at one day, 7 of 12 at two and 6 of 12 at three: an edge that narrows.

B7. Report Table 4.7 in full: best hit rate within an alarm budget

Best hit rate, alarms per year	40	60	80	100
Best model, h = 1	**0.379**	**0.567**	**0.735**	**0.881**
Best model, h = 2	0.366	0.538	0.700	0.840
Best model, h = 3	**0.359**	0.533	0.685	0.827
GRU, 10 seeds, h = 1	0.356–0.377	0.546–0.565	0.719–0.735	0.863–0.881
ARIMA, h = 1	0.358	0.567	0.727	0.867
Gradient boosting (change), h = 1	0.379	0.554	0.727	0.854
Persistence, h = 1	0.371	0.558	0.723	0.873
Climatology, h = 1	0.352	0.521	0.660	0.821
Persistence, h = 3	**0.359**	0.507	0.668	0.791
Climatology, h = 3	0.336	0.510	0.649	0.816

Report Table 4.7. The event is fixed at the health threshold; only the decision threshold varies.

BACKUP

B8. Likely questions, 1 of 2: scope and framing

Why 121 µg/m³?	The CPCB National Air Quality Index boundary of the category below 'Severe', for the 24-hour mean: a policy threshold, not a round number. Section 3.1.
One station: is it representative?	No claim that it is. It is the only nearby monitor with a full decade; CPCB stations serve data only from 19 Feb 2025. A second station is future work. Section 5.3.
Isn't a higher hit rate at a bigger budget trivial?	Yes, by construction, which is why every model comparison is made at a fixed budget. The finding is the size: about 5 points across models against 50 across budgets, and persistence is best at h = 3 and 40 alarms. Section 5.1.
Why does ARIMA beat the ML models?	Consistent with M4: a few thousand observations and no exogenous structure favour a classical model. ML is also more exposed to the uncleaned tail: ridge 0.997 → 1.118. Sections 2.1, 5.3.
Why rMAE rather than RMSE or MASE?	rMAE divides by persistence on identical rows. MASE divides by training volatility and fell for every model on a calm block. DM uses absolute loss to match. Section 3.6.

BACKUP

B9. Likely questions, 2 of 2: evidence and method

Are the network wins real?	Numerically ahead in all ten seeds at h = 1; DM p < 0.05 in 4, 5 and 7 of ten (GRU, LSTM, transformer). None beyond one day. After Bonferroni across 27 tests none of the three survive: read as suggestive. Section 3.7.
Why ten seeds? Why not tune?	The same seed on the same machine moved the GRU almost as much as a new seed, and five to ten overturned the weather result. The GRU was set by inspection; bounded checks say the 3-day setting is the weakest. Section 5.3.
How do you know there is no leakage?	The invariant is tested: corrupt the series from a date, rerun, and no earlier forecast may change. One model per horizon fixes the label arithmetic. Section 3.5.
ARIMA's θ sits near the boundary?	0.013 inside. It crosses at 3 of 28 refits, each with fewer than 1,700 training days; the optimiser is unconstrained and those forecasts are scored as usual. Section 3.3.1.
Would real weather forecasts help?	Untested: only observed weather is used. Weather moved hit rate by at most 1.2 points. Boundary layer height from ERA5 is the stronger candidate. Sections 4.1.1, 6.2.

B10. The words used in this talk, in plain language

Term	In plain words
PM2.5	Fine particles under 2.5 micrometres across, small enough to reach deep into the lungs.
Hazard threshold, 121 µg/m^3	A day above this is 'hazardous': the boundary of the 'Very Poor' category of India's air quality index.
Persistence	The simplest forecast: tomorrow equals today.
ARIMA	A classical time-series model: predicts from recent values and recent errors, with the season added.
Machine learning regressor	A flexible model that learns from examples: ridge regression, random forest, gradient boosting.
Neural network (GRU, LSTM, transformer)	A model that learns patterns from a window of the last 30 days.
rMAE	Average error compared with persistence on the same days: 1.0 is persistence, below 1 is better.
Hit rate	The share of hazardous days that were caught.
False alarm rate	The share of quiet days that were wrongly flagged.
Alarm budget	How many alarms are raised per year: the dial in the talk.
Seed	The random starting point of a network's training; the talk reports ten, as a range.
Rolling origin	Testing as if each forecast were made on its own day, using only the past.